

- Description

This module manages an online shopping discount system, ensuring that:

1. **Each purchase is checked for applicable discounts based on total spending and promotions.**
2. **Discount tiers are applied progressively (e.g., 5% for spending over \$100, 10% for spending over \$200, etc.).**
3. **If a special promo code is provided, an additional discount is applied.**
4. **A log of all purchases, discounts applied, and final prices is maintained for record-keeping.**

This system ensures fair pricing and encourages customers to spend more through progressive discounts.

- Specification

module OnlineShoppingDiscount

var totalAmount, discount, finalPrice: real

var promoCode: string

var purchaseLog: sequence of string

process calculateDiscount(totalAmount: real, promoCode: string) discount: real

functional scenario form:

totalAmount > 200 && discount = totalAmount * 0.10, 10% discount applied.

||

totalAmount > 100 && discount = totalAmount * 0.05, 5% discount applied.

||

promoCode = 20 && discount = discount + 20, Additional \$20 discount applied.

||

promoCode = 10 && discount = discount + 10, Additional \$10 discount applied.

||

discount > totalAmount && discount = totalAmount, Discount capped at total

purchase amount.

end process

end module

- Java Code

```
package code;
import java.util.*;
class Purchase {
    private String customerName;
    private double totalAmount;
    private String promoCode;
    private double discount;
    private double finalPrice;
    public Purchase(String customerName, double totalAmount, String promoCode) {
```

```

        this.customerName = customerName;
        this.totalAmount = totalAmount;
        this.promoCode = promoCode;
        this.discount = calculateDiscount();
        this.finalPrice = totalAmount - discount;
    }

    private double calculateDiscount() {
        double discount = 0.0;
        if (totalAmount > 200) {
            discount = totalAmount * 0.10;
        } else if (totalAmount > 100) {
            discount = totalAmount * 0.05;
        }
        if (promoCode.equalsIgnoreCase("SALE20")) {
            discount += 20.0;
        } else if (promoCode.equalsIgnoreCase("SALE10")) {
            discount += 10.0;
        }
        return Math.min(discount, totalAmount);
    }

    public String getPurchaseDetails() {
        return String.format("Customer: %s | Total: $%.2f | Discount: $%.2f | Final Price: $%.2f | Promo Code: %s",
            customerName, totalAmount, discount, finalPrice, promoCode.isEmpty() ? "None" : promoCode);
    }
}

class OnlineShoppingSystem {
    private List<String> purchaseLog;
    public OnlineShoppingSystem() {
        this.purchaseLog = new ArrayList<>();
    }
    public void processPurchase(String customerName, double amount, String promoCode) {
        Purchase purchase = new Purchase(customerName, amount, promoCode);
        System.out.println(purchase.getPurchaseDetails());
        purchaseLog.add(purchase.getPurchaseDetails());
    }
    public void printPurchaseLog() {
        System.out.println("\nPurchase Log:");
        for (String log : purchaseLog) {
            System.out.println(log);
        }
    }
}

```

```

    }
}
public class c24_OnlineShopping {
    public static void main(String[] args) {
        OnlineShoppingSystem system = new OnlineShoppingSystem();
        // testcase-VT:
        //     system.processPurchase("Alice", 90.0, "");
        //     system.processPurchase("Bob", 120.0, "SALE10");
        //     system.processPurchase("Charlie", 250.0, "SALE20");
        system.processPurchase("David", 50.0, "SALE20");

        // testcase-FT:
        //     system.processPurchase("?", -
1.7976931348606999E308, "");
        //     system.processPurchase("", -2.7430620331195065E303, "");
        //     system.processPurchase("?nsd9", -5.294109726373136E305, "");
        //     system.processPurchase("GGGGW", 2.7233280004857294E293, "");
        // testcase-Z3:
        //     system.processPurchase("Alice", 90.0, "");
        //     system.processPurchase("Bob", 120.0, "SALE10");
        //     system.processPurchase("Charlie", 250.0, "SALE20");
        //     system.processPurchase("David", 50.0, "SALE20");

        // testcase-UVT:
        system.processPurchase("Alice", 90.0, "");
        system.processPurchase("Bob", 120.0, "SALE10");
        system.processPurchase("Charlie", 250.0, "SALE20");
        system.processPurchase("David", 50.0, "SALE20");
        system.printPurchaseLog();
    }
}

```

- Coverage report

1. Test case number: 4
2. V-Method testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
✓ VibrationTestingTool	 1.4 %	12	830	842
✓ src	 1.4 %	12	830	842
✓ code	 1.4 %	12	830	842
✓ c24_OnlineShopping.java	 100.0 %	12	0	12
> c24_OnlineShopping		0	0	0
> OnlineShoppingSystem	 100.0 %	2	0	2
> Purchase	 100.0 %	10	0	10

3. Vibration testing instruction coverage: 98.4%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	1.1 %	180	16,614	16,794
▼ src	1.1 %	180	16,614	16,794
▼ code	1.1 %	180	16,614	16,794
▼ c24_OnlineShopping.java	98.4 %	180	3	183
> c24_OnlineShopping	90.0 %	27	3	30
> OnlineShoppingSystem	100.0 %	45	0	45
> Purchase	100.0 %	108	0	108

4. Fuzz testing branch coverage: 66.7%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.0 %	8	834	842
▼ src	1.0 %	8	834	842
▼ code	1.0 %	8	834	842
▼ c24_OnlineShopping.java	66.7 %	8	4	12
> Purchase	60.0 %	6	4	10
> c24_OnlineShopping		0	0	0
> OnlineShoppingSystem	100.0 %	2	0	2

5. Fuzz testing instruction coverage: 89.6%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	1.0 %	164	16,630	16,794
▼ src	1.0 %	164	16,630	16,794
▼ code	1.0 %	164	16,630	16,794
▼ c24_OnlineShopping.java	89.6 %	164	19	183
> Purchase	85.2 %	92	16	108
> c24_OnlineShopping	90.0 %	27	3	30
> OnlineShoppingSystem	100.0 %	45	0	45

6. Z3 branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.4 %	12	830	842
▼ src	1.4 %	12	830	842
▼ code	1.4 %	12	830	842
▼ c24_OnlineShopping.java	100.0 %	12	0	12
> c24_OnlineShopping		0	0	0
> OnlineShoppingSystem	100.0 %	2	0	2
> Purchase	100.0 %	10	0	10

7. Z3 instruction coverage: 98.4%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	1.1 %	180	16,614	16,794
▼ src	1.1 %	180	16,614	16,794
▼ code	1.1 %	180	16,614	16,794
▼ c24_OnlineShopping.java	98.4 %	180	3	183
> c24_OnlineShopping	90.0 %	27	3	30
> OnlineShoppingSystem	100.0 %	45	0	45
> Purchase	100.0 %	108	0	108

8. Unimproved V-method branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.4 %	12	830	842
▼ src	1.4 %	12	830	842
▼ code	1.4 %	12	830	842
▼ c24_OnlineShopping.java	100.0 %	12	0	12
> c24_OnlineShopping		0	0	0
> OnlineShoppingSystem	100.0 %	2	0	2
> Purchase	100.0 %	10	0	10

9. Unimproved V-method instruction coverage: 98.4%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	1.1 %	180	16,614	16,794
▼ src	1.1 %	180	16,614	16,794
▼ code	1.1 %	180	16,614	16,794
▼ c24_OnlineShopping.java	98.4 %	180	3	183
> c24_OnlineShopping	90.0 %	27	3	30
> OnlineShoppingSystem	100.0 %	45	0	45
> Purchase	100.0 %	108	0	108